HW 7

Monday, 27 February 2023 3:07 PM

Question 5.1

Shell-progressive poisoning ["shrinking core"] model:

Rate of poison deposition: $\frac{d}{dt}\left(C_{ps\infty}\,\frac{4}{3}\pi(R^3-r_c^3)\right)\rho_s = \frac{4}{3}\pi R^3 r_p \cdot \rho_s$

C_{ps}^s = poison conc. at solid surface

C_{ps}^c = " at core boundary

r_c = radius of unpoisoned core

C_p = conc. of poison in bulk

$C_{ps\infty}$ = solid conc. of poison at saturation

$k_{g,p}$ = external interfacial MT coeff.

k'_{sp} = core surface reaction rate coeff. for poison

D_{ep} = effective pore diffusivity of poison

σ_p = adsorption dist. coefficient

S_c = S.A. of core / per unit volume

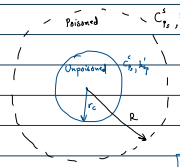
$\langle C_{ps}\rangle$ = pellet average poison conc.

ρ_b = bulk density of catalyst pellet ($\frac{\text{Mass of catalyst}}{\text{Volume}}$)

ε = voidage of packed bed ($\text{porosity} = \frac{\text{Volume of void space}}{\text{volume}}$)

S_a = surf. area / g-cat.

a_c = surf. area / volume.



$\left[\frac{mol}{m^2\cdot s}\right]$ $\left[\frac{m}{s}\right]$ $\left[\frac{mol}{m^3}\right]$

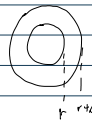
External MT limitations: Flux $= k_{gp}(C_p - C_{ps}^s)$

Rate of mass transport to the external surface = Flux · Area

$= (k_{gp}(C_p - C_{ps}^s))\,(4\pi R^2)$

$\left[\frac{mol}{s}\right] = \frac{mol}{m^2\cdot s}\cdot m^2 \cdot \frac{g}{g}\cdot$

Steady state diffusion through a spherical shell:



$W\cdot 4\pi r^2|_r - W\cdot 4\pi r^2|_{r+\Delta r} = 0$ (Diffusion within)

$\frac{d(W\cdot 4\pi r^2)}{dr} = 0$; $W = -D_e\frac{dC}{dr}$

$\frac{d}{dr}\left(D_e\cdot\frac{dc}{dr}\cdot r^2\right) = 0$

$D_e\frac{d^2c}{dr^2}r^2 + D_e 2r\cdot\frac{dc}{dr} = 0$

$\Rightarrow D_e\frac{d^2c}{dr^2} + \frac{2}{r}\cdot D_e\frac{dc}{dr} = 0$

general solution $\rightarrow C = \frac{c_1}{r} + c_2$

BC: 1) @ $r = r_c$, $C_{ps}^c = C$

2) @ $r = R$, $C = C_{ps}^s$

$C_{ps}^c = \frac{c_1}{r_c} + c_2 \Rightarrow c_2 = C_{ps}^c - \frac{c_1}{r_c}$

$C_{ps}^s = \frac{c_1}{R} + c_2$

$(C_{ps}^s - C_{ps}^c) = c_1\left(\frac{1}{R} - \frac{1}{r_c}\right)$

$c_1 = \frac{(C_{ps}^s - C_{ps}^c)}{\left(\frac{1}{R} - \frac{1}{r_c}\right)} \Rightarrow c_2 = C_{ps}^c - \frac{1}{r_c}\frac{(C_{ps}^s - C_{ps}^c)}{\left(\frac{1}{R} - \frac{1}{r_c}\right)}$

$c_2 = C_{ps}^c - \frac{(C_{ps}^s - C_{ps}^c)}{\left(\frac{r_c}{R} - 1\right)}$

Fick's law diffusion (Flux): $\left(-D_e\frac{dc}{dr}\right)\cdot$ Area

$= +D_e\left(\frac{c_1}{r^2}\right)\cdot 4\pi r^2$

$= \frac{D_e\cdot(C_{ps}^s - C_{ps}^c)}{\left(\frac{1}{R} - \frac{1}{r_c}\right)}\cdot 4\pi$

Deposition rate at boundary: Transport of species = Reaction at the surface

$4\pi r_c^2 k_{sp} C_{ps}^c = 4\pi r_c^2 \cdot k'_{sp}\cdot C_{ps}^c$

Defining $\langle C_{ps}\rangle$ as:

$\frac{\langle C_{ps}\rangle}{C_{ps\infty}} = 1 - \left(\frac{r_c}{R}\right)^3$; $\xi = \frac{r_c}{R}$ $\Rightarrow \frac{d}{dt}\left(\frac{\langle C_{ps}\rangle}{C_{ps\infty}}\right) = -\frac{3}{R}r_c^2\cdot\frac{dr_c}{dt}$

$$C_{pl\infty} \cdot \frac{4}{3}\pi \frac{d}{dt}\left(R^3 - r_c^3\right) \cdot$$

$$= \rho_s C_{pl\infty} \frac{4}{3}\pi \left(-3 \frac{dr_c}{dt}\right)$$

$$\frac{4}{3}\pi R^3 r_p \rho_s = 4\pi R^2 \rho_s C_{pl\infty} \frac{dr_c}{dt}$$

$$\frac{dr_c}{dt} = \frac{3R^2 r_p}{C_{pl\infty}}$$

A B

1) $$\left(\frac{4}{3}\pi R^3 \rho_s\right) \cdot r_p = \left(4\pi R^2 k_{gp} \rho_s\right)\left(C_p - C_{ps}^s\right)$$

$$C_p = \frac{A}{B} r_p + C_{ps}^s$$

C

2) $$A r_p = \left(4\pi r_c^2 k_{sp}\right) C_{ps}^c$$

$$C_{ps}^c = \frac{A}{C} r_p$$

D

3) $$A r_p = \left(\frac{4\pi D_{ap} \rho_s}{\left(\frac{1}{r_c} - \frac{1}{R}\right)}\right)\left(C_{ps}^s - C_{ps}^c\right)$$

$$A r_p = D\left(C_{ps}^s - C_{ps}^c\right)$$

$$A r_p = D C_{ps}^s - \frac{D \cdot A}{C} r_p$$

$$C_{ps}^s = \left(\frac{A}{D} + \frac{A}{C}\right) r_p$$

$$\therefore \quad C_p = \frac{A}{B} r_p + \left(\frac{A}{D} + \frac{A}{C}\right) r_p$$

$$C_p = A \cdot r_p \left(\frac{1}{B} + \frac{1}{D} + \frac{1}{C}\right)$$

$$A \cdot r_p = \frac{C_p}{\left(\frac{1}{B} + \frac{1}{D} + \frac{1}{C}\right)}$$

4) We also know,

$$A \cdot r_p = \frac{d}{dt}\left[C_{pl\infty} \frac{4\pi}{3}\left(R^3 - r_c^3\right)\right] \cdot \rho_s$$

$$\frac{C_p}{\left(\frac{1}{B} + \frac{1}{D} + \frac{1}{C}\right)} = \frac{4}{3}\pi \rho_s \frac{d}{dt}\left[C_{pl\infty} \cdot \left(R^3 - r_c^3\right)\right] \quad ; \quad \frac{\langle C_{pl}\rangle}{C_{pl\infty}} = \frac{R^3 - r_c^3}{R^3}$$

$$R^3 \cdot \langle C_{pl}\rangle = C_{pl\infty}\left(R^3 - r_c^3\right)$$

$$\frac{C_p}{\left(\frac{1}{B} + \frac{1}{D} + \frac{1}{C}\right)} = \frac{4}{3}\pi \rho_s \frac{d}{dt}\left[R^3 \cdot \langle C_{pl}\rangle\right]$$

$$\Rightarrow \quad \frac{3 C_p}{4\pi \rho_s R^3 \left(\underbrace{\frac{1}{4\pi R^2 k_{gp} \rho_s}}_{\text{Sh term}} + \underbrace{\frac{1}{4\pi r_c^2 k_g}}_{\text{Da term}} + \underbrace{\frac{1}{4\pi D_{ap} \rho_s \left(\frac{1}{r_c} - \frac{1}{R}\right)}}_{\text{E term}}\right)} = \frac{d}{dt}\langle C_{pl}\rangle$$

$$\Rightarrow \quad \frac{3 C_p}{\rho_s \cdot D_e \cdot R^3 \cdot \left(\frac{1}{R \cdot \left(\frac{2 k_{gp} R}{D_e}\right)} + \frac{1}{\left(\frac{k_{sp} \cdot R}{D_{ap}}\right)\frac{r_c^2}{R}} + \frac{1}{\rho_s\left(\frac{1}{r_c} - \frac{1}{R}\right)}\right)} = \frac{d}{dt}\langle C_{pl}\rangle$$

$$\Rightarrow \frac{3D_e C_p}{\rho_s R^3}\left(\frac{1}{\frac{1}{R\cdot Sh_p}+\frac{1}{D_a\cdot\frac{r_c^2}{\varepsilon_c R}}+\frac{1}{\rho_s\left(\frac{1}{r_c}-\frac{1}{R}\right)}}\right)=\frac{d}{dt}\langle C_{pR}\rangle$$

Taking $\frac{1}{R}$ as factor

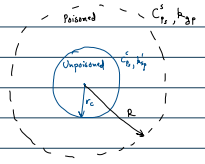
$$\Rightarrow \frac{3D_e C_p}{\rho_s R^2\frac{1}{R}}\left(\frac{1}{\frac{1}{Sh_p}+\frac{1}{D_a\left(\frac{r_c}{R}\right)^2}+\frac{1}{\rho_s\left(\frac{R-r_c}{R\cdot c}\right)}}\right)=\frac{d}{dt}\langle C_{lp}\rangle$$

Multiplying by t_{ref} expressions & $C_{pl\infty}$ to get:

$$\boxed{\frac{3D_e\cdot t_{ref}\,C_{p,ref}}{R^2\cdot\rho_s\cdot C_{pl\infty}}\left(\frac{1}{\frac{1}{Sh_p}+\frac{(1-\varepsilon_c)}{\varepsilon_c}+\frac{1}{D_a\varepsilon_c^2}}\right)=\frac{d}{dt'}\left[\frac{\langle C_{lp}\rangle}{C_{pl\infty}}\right]}$$

Question 2 (5-2)

+ Poisoned shell is completely inactive
+ First-order reaction (in unpoisoned core)



Diffusion in poisoned shell: $D'_{eA}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dC'_{As}}{dr}\right)=0 \qquad r_c<r<R \quad \ldots (1)$

Diffusion to pellet core is equal to the reaction rate: $D_{eA}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dC_{As}}{dr}\right)=k_{rA}\rho_s\cdot C_{As} \qquad 0<r<r_c \quad \ldots (2)$

BC's: @ $r=R$, $D'_{eA}\frac{dC'_{As}}{dr}=k_{gA}(C_A-C^s_{As})$ [External MT limitations]

@ $r=r_c$, $C_{As}=C'_{As}$; $D_{eA}\frac{dC_{As}}{dr}=D'_{eA}\frac{dC'_{As}}{dr}$ [Symmetry BC]

} pseudo-steady state

@ $r=0$, C_{As} = finite value

Differential equations:

(1) $\frac{d}{dr}\left(r^2\frac{dC'_{As}}{dr}\right)=0$

$r^2\frac{d}{dr}C'_{As}=C_1$

$\frac{d}{dr}C'_{As}=\frac{C_1}{r^2}$

$C'_{As}=-\frac{C_1}{r}+C_2$

@ $r=r_c$:

$C'_{As}=-\frac{C_1}{r_c}+C_2$

@ $r=R$;

$\frac{dC'_{As}}{dr}=\frac{k_{gA}}{D'_{eA}}(C_A-C^s_{As})$

$\frac{C_1}{R^2}=\frac{k_{gA}}{D'_{eA}}(C_A-C^s_{As})$

$C_1=\frac{k_{gA}R^2}{D'_{eA}}(C_A-C^s_{As})$

$C_2=C'_{As}+\frac{C_1}{r_c}=C'_{As}+\frac{k_g R^2}{D'_{eA}r_c}(C_A-C^s_{As})$

$$\therefore \cancel{C_{AS}} = \frac{k_{gA} R^2}{D_{eA}} \left(-\frac{1}{r}(C_A - C_{AS}^s) + \frac{1}{r_c}(C_A - C_{AS}^s) \right) + \cancel{C_{AS}}$$

(2) $$\frac{d}{dr}\left(r^2 \frac{dC_{AS}}{dr} \right) = \frac{k_{rA} \rho_s C_{AS} \cdot r^2}{D_{eA}}$$

$$\Rightarrow r^2 \frac{d^2 C_{AS}}{dr^2} + 2r \cdot \frac{dC_{AS}}{dr} - \underbrace{\frac{k_{rA} \rho_s}{D_{eA}}}_{a} \cdot C_{AS} \cdot r^2 = 0$$

$$C_{AS} = \frac{c_1}{r} e^{-\sqrt{a} \cdot r} + \frac{c_2}{\sqrt{a} \cdot r} e^{\sqrt{a} \cdot r}$$

@ $r = R$,
$$D_e' \frac{dC_{AS}'}{dr} = k_{gA}(C_A - C_{AS}^s)$$

@ $r = r_c$,
$$C_{AS} = C_{AS}' \quad ; \quad D_e \frac{dC_{AS}}{dr} = D_e' \frac{dC_{AS}'}{dr}$$